

# SWE-EVO: Benchmarking Coding Agents in Long-Horizon Software Evolution Scenarios

Tue Le<sup>1,\*</sup> Minh Vu Thai Pham<sup>1,\*</sup> Dung Nguyen Manh<sup>2</sup> Huy Nhat Phan<sup>1</sup> Nghi D. Q. Bui<sup>3,†</sup>

<sup>1</sup>FPT Software AI Center, Ha Noi, Viet Nam

<sup>2</sup>School of Computing and Information Systems, University of Melbourne

<sup>3</sup>Center of AI Research, VinUniversity, Ha Noi, Viet Nam

Existing benchmarks for AI coding agents focus on isolated, single-issue tasks such as fixing a bug or adding a small feature. However, real-world software engineering is a long-horizon endeavor: developers interpret high-level requirements, coordinate changes across many files, and evolve codebases over multiple iterations while preserving functionality. We introduce SWE-EVO, a benchmark for this long-horizon software evolution challenge. Constructed from release transitions in seven mature open-source Python projects, SWE-EVO comprises 48 release-sized tasks requiring multi-step modifications spanning an average of 21 files, validated against test suites averaging 874 tests per instance. Experiments reveal a striking capability gap: the best model reaches only 25% on SWE-EVO, while gpt-5.2 drops from 72.80% on SWE-Bench Verified to 22.92% on SWE-EVO, showing that current agents struggle with sustained, multi-file reasoning. We also propose *Fix Rate*, a metric capturing partial progress on these complex, long-horizon tasks.

 <https://github.com/SWE-EVO/SWE-EVO>

## 1. Introduction

Large language models (LLMs) have achieved remarkable progress in automating software engineering (SE) tasks, including code generation (Bui et al., 2023; Chen et al., 2021a; Li et al., 2022; Manh et al., 2023; To et al., 2023; Wang et al., 2023; Wei et al., 2023; Zhuo et al., 2024), bug fixing (Jimenez et al., 2023; Xia et al., 2024), and test synthesis (Chen et al., 2022; Jain et al., 2024; Wang et al., 2024b). These advancements have facilitated the emergence of AI-powered coding agents capable of assisting or automating key aspects of the software development lifecycle (Fan et al., 2023; Gao et al., 2025; He et al., 2025; Zhang et al., 2023).

Building on these capabilities, multi-agent systems have rapidly emerged to address long-horizon challenges in software engineering. By coordinating specialized agents for navigation, localization, patching, and verification, these frameworks improve scalability and performance over single-agent architectures. Recent systems (Nguyen et al., 2025b; Phan et al., 2024; Wang et al., 2024d; Yang et al., 2024a) illustrate this shift in real-world repositories. Industry adoption reflects this momentum: over 90% of engineering teams now integrate generative AI into SE workflows, up from 61% in 2024 (DORA Research Program, 2025).

To evaluate these agents rigorously, benchmarks have become essential. Early efforts like HumanEval (Chen et al., 2021b) focused on function-level code completion (Chen et al., 2021a),

\*Equal contribution      †Project lead

Correspondence to: Tue Le <tueldt1@fpt.com>, Minh Vu Thai Pham <minhpvt@fpt.com>, Dung Nguyen Manh <manhdung.nguyen.1@student.unimelb.edu.au>, Huy Nhat Phan <huyphn168@gmail.com>, Nghi D. Q. Bui <bdqnghi@gmail.com>.

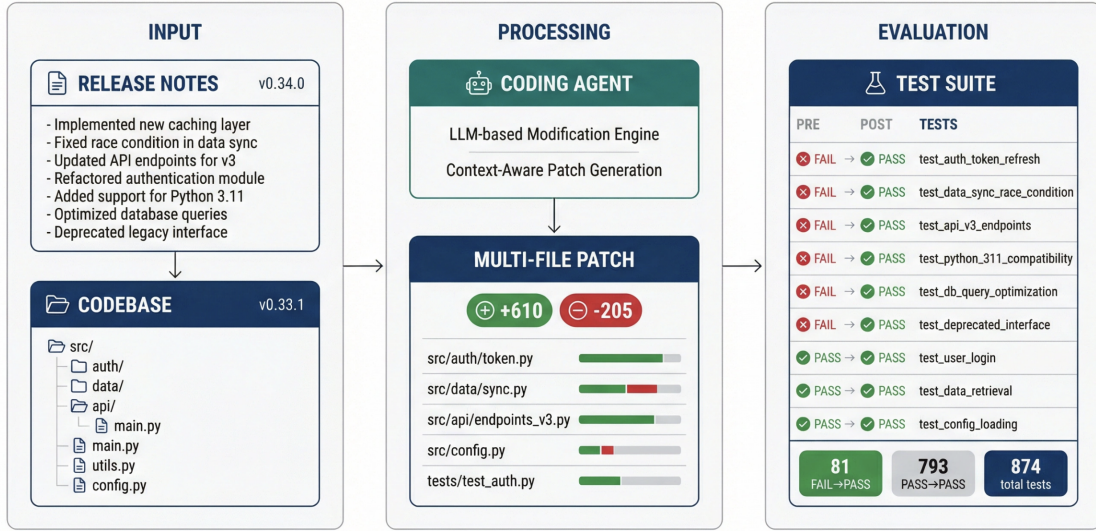


Figure 1 | Overview of the SWE-EVO task pipeline. **Input:** the agent receives release notes specifying multiple changes (bug fixes, new features, maintenance) alongside the full codebase at the start version. **Processing:** a coding agent generates a multi-file patch spanning dozens of files. **Evaluation:** the patch is validated against a comprehensive test suite comprising FAIL→PASS tests (verifying the required changes) and PASS→PASS tests (ensuring no regressions).

while SWE-Bench (Jimenez et al., 2023) curated real-world GitHub issues requiring verifiable patches for isolated problems (Jimenez et al., 2023). SWE-Bench is now a de facto standard for assessing multi-agent capabilities in practical coding scenarios.

While recent models reach around 75% on SWE-Bench-Verified (e.g., GPT-5.2 (OpenAI, 2025c)) and around 40% on the full leaderboard (e.g., OpenCSG Starship at 39.67% (Jimenez et al., 2024)), the benchmark shows signs of saturation on isolated tasks. More importantly, SWE-Bench focuses on discrete issue resolution and does not capture the core SE challenge: continuous evolution of existing systems (Kaur and Singh, 2015; Singh et al., 2019). In practice, up to 80% of effort is spent maintaining and evolving legacy code, requiring coordinated changes across modules, versions, and specifications (Kaur and Singh, 2015; Singh et al., 2019).

This gap between benchmark tasks and real-world evolution scenarios motivates our central research question: *Given an existing codebase, can multi-agent LLM systems autonomously evolve the system in response to dynamic input requirements, demonstrating sustained planning, adaptability, and innovation across long-horizon tasks?*

To address this gap, we introduce SWE-EVO, a benchmark for autonomous software evolution rather than single-issue repair. SWE-EVO uses release notes, commit histories, and versioned snapshots from mature open-source Python projects to construct tasks where agents interpret Software Requirement Specifications (SRS), plan multi-step modifications, and evolve codebases across releases. SWE-EVO comprises 48 release-sized tasks across 7 repositories (Table 1 and Figure 3). We describe construction in Section 3 and results in Section 4.

Figure 1 illustrates the SWE-EVO pipeline: from release notes and a start-version codebase, the agent generates a multi-file patch validated by comprehensive tests. Software evolution (Figure 7 in Appendix B) requires SRS alignment, holistic program understanding, and coordinated changes, unlike SWE-Bench’s isolated issue setting (Figure 2).

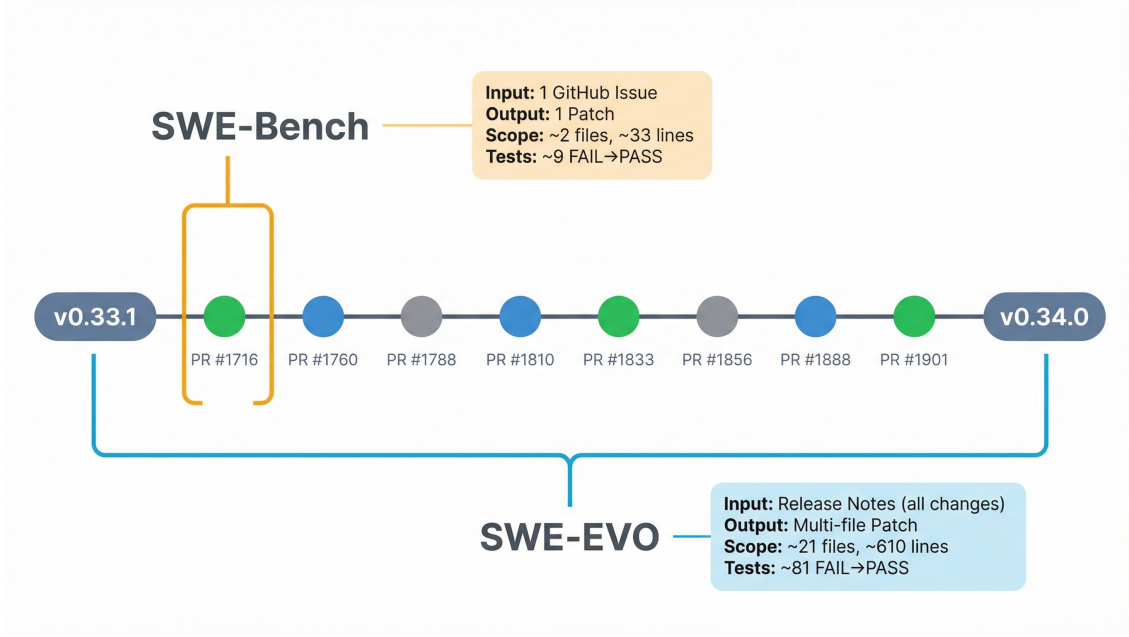


Figure 2 | Comparison of SWE-Bench and SWE-EVO. SWE-Bench tasks agents with resolving a single GitHub issue to produce one patch. SWE-EVO requires agents to interpret release notes and implement comprehensive changes that resolve multiple PRs, develop new features, and fix multiple bugs to evolve the codebase to a new version.

Our evaluation with OpenHands, SWE-agent, and 18 models reveals a significant capability gap: the best model (gpt-5.4) resolves only 25% of SWE-EVO, while gpt-5.2 drops from 72.80% on SWE-Bench Verified to 22.92%. Larger models generally outperform smaller variants. Failure analysis suggests that stronger models more often misinterpret nuanced release notes, while weaker models struggle with tool use and syntax errors.

## 2. Related Work

### 2.1. Code Generation and SE Benchmarks

Code generation evaluation has moved from function-level tasks (Austin et al., 2021; Chen et al., 2021a; Hendrycks et al., 2021) to repository-level SE challenges. Early benchmarks such as HumanEval, MBPP, and APPS remain largely single-file and increasingly saturated (Zhou et al., 2023), while newer efforts such as CodeMMLU (Nguyen et al., 2025a) and CodeWiki (Hoang et al., 2025) broaden evaluation to code understanding and documentation. SWE-bench (Jimenez et al., 2024) shifted the field to verifiable patches for real GitHub issues, with curated subsets (OpenAI, 2024b) and extensions across languages (Zan et al., 2025), multimodal inputs (Yang et al., 2024b), live collection (Badertdinov et al., 2025; Zhang et al., 2025b), and enterprise-scale settings (Scale AI, 2025). EnConda-Bench (Kuang et al., 2025) isolates environment-configuration trajectories, including setup planning, diagnosis, repair, and execution. SWE-EVO instead targets version-to-version software evolution, where agents must implement release-level changes in the codebase rather than fix injected configuration faults. Prior benchmarks therefore emphasize isolated issue resolution or narrow subskills more than multi-step evolution across commits and longer planning horizons.

## 2.2. Software Engineering Agents

Interactive bug-fixing work (Chen et al., 2023; Xia and Zhang, 2023, 2024; Yuan et al., 2024) and autonomous systems such as Devin AI (Cognition, 2024) established the basis for SE agents. SWE-agent (Yang et al., 2024a) emphasized the agent-computer interface; OpenHands (Wang et al., 2024d) presented a multi-agent platform evaluated on 15 benchmarks; AutoCodeRover (Zhang et al., 2024) paired LLMs with AST-based search; and Agentless (Xia et al., 2024) showed that a simple localization-repair pipeline can remain competitive. Multi-agent designs further explore role specialization and coordination, including AgentCoder (Huang et al., 2023), AgileCoder (Nguyen et al., 2025b), and CodeAct (Wang et al., 2024c). Model-side progress includes post-training on real SE data (Carbonneaux et al., 2025; Chen et al., 2025; DeepSeek AI, 2025; Kimi Team, 2025b; Luo et al., 2025; Wei et al., 2025) and synthetic-data generation (Pham et al., 2025). A related line studies self-evolving agents and tool construction (Cai et al., 2024; Qian et al., 2024; Qiu et al., 2025; Robeyns et al., 2025; Wang et al., 2024a, 2025, 2024e; Zhang et al., 2025a) (Appendix C).

## 2.3. Context Engineering for Long-Horizon Agents

Long-horizon agents depend on effective context management, which is central to SWE-EVO because tasks unfold across large codebases and many turns. Surveys (Hua et al., 2025; Mei et al., 2025) frame the area around retrieval, processing, and management, tracing a path from prompt engineering and RAG to context engineering. Representative methods include compression (Ge et al., 2024), retrieval schemes such as Self-RAG (Asai et al., 2024), RAPTOR (Sarthi et al., 2024), and GraphRAG (Gutierrez et al., 2024), and memory systems such as MemGPT (Packer et al., 2023) and hierarchical storage (Zhong et al., 2024) that extend beyond single sessions. Meta Context Engineering (Ye et al., 2026) pushes this further by optimizing context assembly, reporting 89.1% on SWE-bench Verified versus 70.7% for hand-engineered baselines. These ideas are directly relevant to SWE-EVO, where agents must reason over long specifications and patches spanning many files.

# 3. SWE-EVO Dataset

SWE-EVO is a benchmark constructed from release notes and version histories of popular open-source repositories, capturing real software evolution scenarios where coding agents must interpret high-level software requirement specifications, plan and implement multi-step modifications across versions, and ensure that the evolved system passes validation tests aligned with those specifications.

## 3.1. Benchmark Construction

The construction of SWE-EVO consists of three major phases: (1) repository selection and data scraping, (2) candidate selection and filtering, and (3) execution-based filtering. We describe each phase in turn.

**Stage I: Repository Selection and Data Scraping.** To maximize reproducibility and comparability, we inherit repositories and execution environments from SWE-bench and it keeps our benchmark “plug-and-play” for existing SWE agents. Concretely, we begin by collecting samples from SWE-bench (Jimenez et al., 2024) and from SWE-gym (Pan et al., 2024) as our seed pool of task instances. Both resources already give us: a real repository snapshot, an executable environment, and a set of tests tied to a human-authored change.

**Stage II: Candidate Selection and Filtering.** Unlike SWE-Bench, which frames tasks as resolving a single issue, we target evolving a codebase between two release versions. We therefore create



|            |                    | Mean   | Max   |
|------------|--------------------|--------|-------|
| Issue Text | Length (Words)     | 2390.5 | 22344 |
| Codebase   | # Files (non-test) | 363    | 1046  |
|            | # Lines (non-test) | 78K    | 272K  |
| Gold Patch | # Lines edited     | 610.5  | 4113  |
|            | # Files edited     | 20.9   | 105   |
|            | # Func. edited     | 51.0   | 379   |
| Tests      | # Fail to Pass     | 81.4   | 2774  |
|            | # Total            | 874.0  | 8552  |

Table 1 | Average and maximum numbers characterizing different attributes of a SWE-EVO task instance. Statistics are micro-averages calculated without grouping by repository.

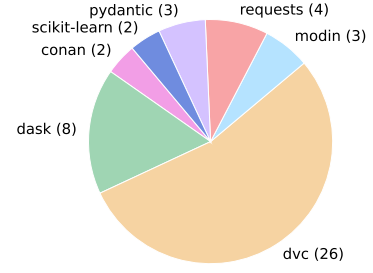


Figure 3 | Distribution of SWE-EVO tasks (in parenthesis) across 7 open source GitHub repositories that each contains the source code for a popular, widely downloaded PyPI package.

candidate instances by selecting only those samples whose base commit corresponds exactly to a version tag of the repository (i.e., a release snapshot). For each such candidate, we define the problem statement as the release-note delta between that version and its next tagged version; the agent’s job is to implement the specified changes across the codebase. This framing treats a release as the unit of evolution, matching how users and maintainers typically receive a mixture of fixes, features, and maintenance changes in open-source projects. Importantly, SWE-EVO is not intended to imitate a single clean pull request or one atomic commit. Instead, it evaluates whether an agent can coordinate the full release-level behavioral delta. This formulation intentionally withholds an oracle decomposition into a sequence of separately specified PRs: agents may still work iteratively during their trajectory, but evaluation is based on the final evolved repository state. A sequential PR-by-PR variant is a useful future setting, but it would provide cleaner intermediate supervision and move closer to repeated single-issue repair.

**Stage III: Execution-Based Filtering.** Following SWE-Bench, we validate each candidate by applying the instance’s test patch content and logging test outcomes *before* and *after* the remaining patch content is applied. We retain only instances that exhibit at least one FAIL\_TO\_PASS test (i.e., a test that fails pre-patch and passes post-patch), ensuring a measurable behavioral change attributable to the required evolution. We additionally discard candidates that trigger installation or runtime errors under the benchmark environment. The resulting set comprises evolution tasks with verifiable behavioral deltas and stable execution characteristics.

After the automatic filtering steps yielded a few hundred candidate release transitions, we used human effort to manually verify task quality and scale the benchmark down to 48 high-confidence instances. This quality-first pass checks alignment among the release note, linked PR/issue context, gold patch, and executable tests. Figure 3 reports their distribution across repositories, while Table 1 summarizes key characteristics. Compared with SWE-Bench, SWE-EVO has longer specifications, broader patches, and heavier test suites (Figure 4), making each human-curated release transition substantially larger in engineering scope than a single-issue repair task. We therefore treat benchmark scale along two axes: number of instances and scope per instance. SWE-EVO is smaller on the former, but much larger on the latter, so we avoid over-interpreting close leaderboard gaps while using these release-sized tasks as a stress test of software-evolution capability.

Figure 4 makes this scale difference explicit. Across specification length, edited lines, edited files, edited functions, FAIL\_TO\_PASS tests, and total tests, SWE-EVO is consistently larger than SWE-

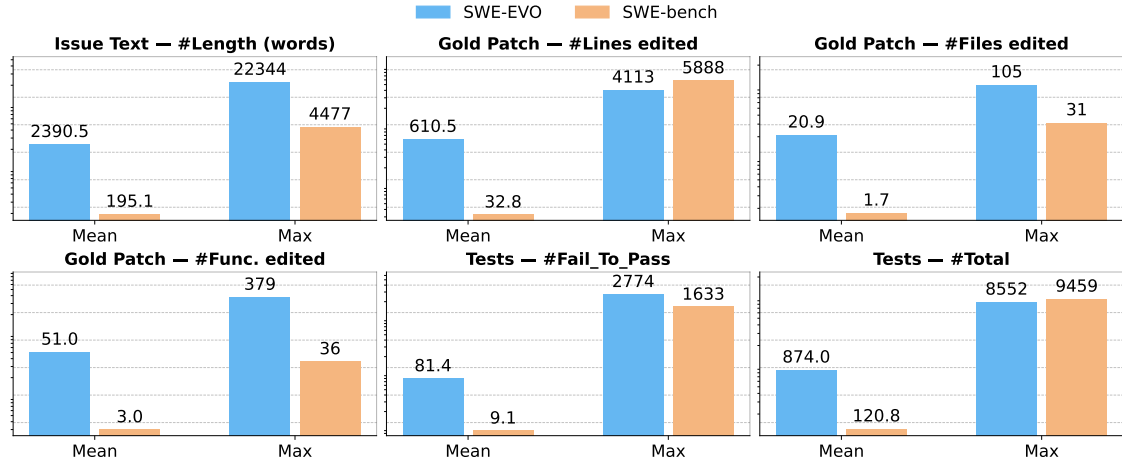


Figure 4 | SWE-EVO is markedly more demanding than SWE-Bench, with longer specifications, broader patch scope, and heavier test suites in both mean and max.

Bench in both average and worst-case size. These differences explain why SWE-EVO stresses release-level software evolution: agents must interpret broader requirements, coordinate changes across more code, and satisfy substantially heavier verification suites.

### 3.2. Task Formulation

Having described how SWE-EVO instances are constructed, we now formalize the task definition and evaluation metrics.

**Model input.** A model is provided with (i) a release-note-centered specification of the intended change (bug fix or feature refinement), consisting of the release note itself and, in our default setting, the text of any PRs or issues explicitly linked from that release note (Appendix N.1), and (ii) a complete codebase at the *pre-release* commit. The model must produce edits to the codebase that implement the described change. In practice, we represent a solution as a patch file that specifies which lines to modify across files. For clarity, we refer to the patch generated by the model as the *model patch*, while the ground-truth patch extracted from the start-version to end-version change inside the instance is referred to as the *gold patch*.

**Evaluation Metrics.** Consistent with the evaluation setup of SWE-bench (Jimenez et al., 2024), we use the **Resolved Rate (%)** as the primary metric, representing the proportion of task instances successfully solved by the agent. In addition, we report the **Patch Apply Rate (%)**, which measures the percentage of generated patches that are syntactically valid and can be applied to the codebase without errors. Beyond the hard-score **Resolved Rate (%)**, we introduce a soft-score, **Fix Rate (%)**, to provide a more fine-grained assessment of SWE-EVO while remaining consistent with Resolved Rate.

*Resolved Rate.* Following SWE-Bench, this metric focuses on two categories in the test outcomes:

- **FAIL\_TO\_PASS** tests that were initially failing (FAILED or ERROR) but pass (PASSED after applying the gold patch).

- **PASS\_TO\_PASS** tests pass both before and after the gold patch and serve as regression checks to ensure unrelated functionality is preserved.

The **Resolved Rate** of an instance is equal 1 if *all* tests in both **FAIL\_TO\_PASS** and **PASS\_TO\_PASS** are PASSED; otherwise, it is 0. Hence, this metric imposes a strict binary criterion: an instance is counted as resolved only if every relevant test succeeds.

*Fix Rate: A Soft Metric.* While Resolved Rate provides a clear pass/fail signal, it may hide meaningful partial progress, especially when instances contain thousands of tests (Table 1, Figure 4). In such cases, models may fix many failing tests without fully resolving the instance. To capture this progress, we introduce **Fix Rate**, a soft metric that measures the fraction of **FAIL\_TO\_PASS** tests successfully fixed.

Fix Rate also enforces a regression constraint: if any **PASS\_TO\_PASS** test fails after applying the patch, the instance receives a score of 0. This rewards partial fixes while penalizing regressions. We use this constraint because a valid evolution should implement the requested change without breaking existing behavior. Let  $F_i$  denote the **FAIL\_TO\_PASS** tests and  $P_i$  the **PASS\_TO\_PASS** tests for instance  $i$ . The **Fix Rate** is defined in Equation 1 as:

$$\text{Fix Rate}(i) = \begin{cases} \frac{|\{t \in F_i \mid t \text{ passes}\}|}{|F_i|}, & \text{if all } t \in P_i \text{ pass;} \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

The overall **Fix Rate (%)** across all  $N$  instances is  $100 \times \frac{1}{N} \sum_{i=1}^N \text{Fix Rate}(i)$ .

Fix Rate lies in  $[0, 1]$  and is consistent with Resolved Rate: an instance is resolved when its Fix Rate equals 1. Aggregated across instances, this metric captures partial progress that binary Resolved Rate cannot reflect, while remaining aligned with it (Table 6). We keep strict Fix Rate as the main soft metric because valid software evolution should make progress without regressing existing behavior. For diagnostics, the same test logs can also report relaxed **FAIL\_TO\_PASS** progress and **PASS\_TO\_PASS** preservation separately, distinguishing incomplete implementations from regressions (Appendix J). Fix Rate is therefore an execution-based progress signal, not a complete measure of code quality, maintainability, or patch minimality.

### 3.3. Features of SWE-EVO

SWE-EVO differs from SWE-Bench along several key dimensions. First, it presents substantially richer supervision: longer specifications, broader patches (more lines, files, and functions edited), and heavier test suites (Figure 4). Second, unlike SWE-Bench where each task maps to a single pull request, SWE-EVO instances may aggregate multiple PRs, yielding a wide range of difficulty levels (see Appendix F). Finally, evaluation is robust: at least one **FAIL\_TO\_PASS** test validates each gold patch (81% have two or more), with an additional mean of 793 **PASS\_TO\_PASS** regression tests per instance.

## 4. Experiments

### 4.1. Experiment Setup

We evaluate SWE-EVO with two coding-agent scaffolds and a diverse model suite covering frontier proprietary systems and open-weight models.

**Agents and Model Selection.** We evaluate SWE-EVO using two established coding agent frameworks: **OpenHands** (Wang et al., 2024d) (with CodeActAgent, max 100 iterations) and **SWE-agent** (Yang et al., 2024a) (max 100 LLM calls). We test 18 state-of-the-art LLMs spanning five providers: OpenAI (10 models including GPT-5 series, o3, GPT-4.1, GPT-4o, and GPT-oss-120b), DeepSeek (3 models), Zhipu AI (3 models), Qwen (1 model), and Moonshot AI (2 models). The full list of models with references is provided in Table 3 in the Appendix.

For all reasoning models, we use a medium reasoning effort setting, which balances inference cost and accuracy.

## 4.2. Performance on SWE-EVO

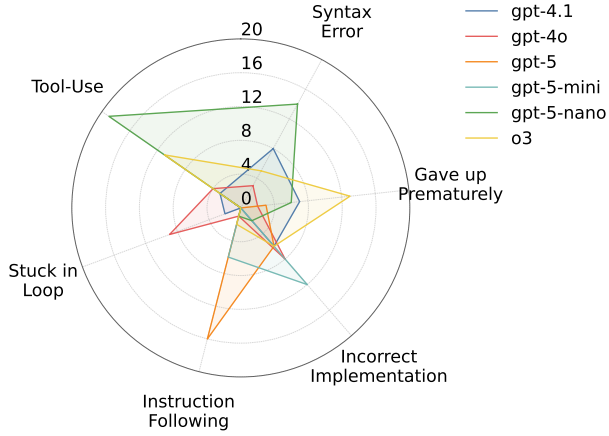
Each SWE-EVO task is specified by a release-note entry, optionally augmented with linked PR/issue text. To prevent models from retrieving future code or online PR content during evaluation, we block internet access and provide the linked PR/issue text directly in our default *release-note* + *PR/issue context* setting (Appendix N.1). Table 2 reports results for OpenHands and SWE-agent, alongside each model’s SWE-bench Verified score when available.

| Model       |                       | SWE-EVO<br>(OpenHands) |         | SWE-EVO<br>(SWE-agent) |         | SWE-Bench Verified<br>(bash only) |
|-------------|-----------------------|------------------------|---------|------------------------|---------|-----------------------------------|
|             |                       | % Resolved             | % Apply | % Resolved             | % Apply | % Resolved <sup>†</sup>           |
| OpenAI      | gpt-5.4               | 25.00                  | 97.92   | 25.00                  | 97.92   | —                                 |
|             | gpt-5.2               | 18.75                  | 100     | 22.92                  | 97.92   | 72.80                             |
|             | gpt-5-08-07           | 18.75                  | 100     | 20.83                  | 100     | 65.00                             |
|             | gpt-5-mini-08-07      | 10.42                  | 97.92   | 10.42                  | 100     | 59.80                             |
|             | gpt-5-nano-08-07      | 4.17                   | 85.42   | 4.17                   | 100     | 34.80                             |
|             | o3-2025-04-16         | 4.17                   | 93.75   | 6.25                   | 100     | 58.40                             |
|             | gpt-4.1-2025-04-14    | 2.08                   | 87.50   | 10.42                  | 97.92   | 39.58                             |
|             | gpt-4o-2024-11-20     | 6.25                   | 97.92   | 6.25                   | 100     | 21.62                             |
|             | gpt-oss-120b          | 2.08                   | 100     | 6.25                   | 100     | 26.00                             |
|             |                       |                        |         |                        |         |                                   |
| DeepSeek    | deepseek-v3p2         | 20.83                  | 95.83   | 23.40                  | 95.83   | 70.00                             |
|             | deepseek-v3p1         | 16.67                  | 97.92   | 10.42                  | 100     | —                                 |
|             | deepseek-r1-0528      | 10.42                  | 100     | 8.33                   | 100     | 57.60                             |
| Zhipu AI    | glm-5                 | 8.33                   | 97.92   | 37.50                  | 100     | 72.80                             |
|             | glm-4p7               | 4.17                   | 100     | 39.58                  | 97.92   | —                                 |
|             | glm-4p5               | 16.67                  | 97.92   | 16.67                  | 100     | 54.20                             |
| Qwen        | qwen3-coder-480b-a35b | 14.58                  | 97.92   | 14.58                  | 97.92   | 55.40                             |
| Moonshot AI | kimi-k2p5             | 22.92                  | 97.92   | 25.00                  | 97.92   | 70.80                             |
|             | kimi-k2-instruct      | 16.67                  | 100     | 18.75                  | 100     | 43.80                             |

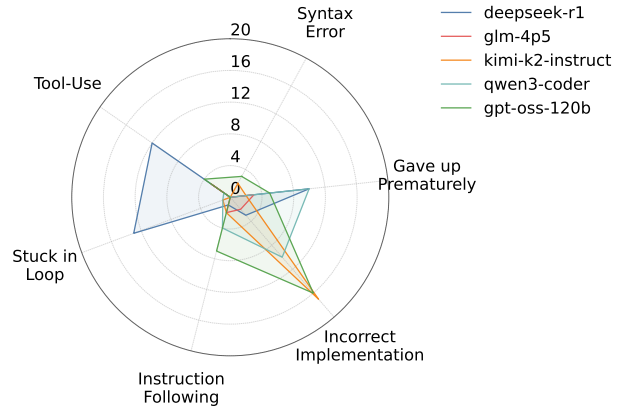
Table 2 | Results on SWE-EVO with *release note* + *PR/issue context*: Resolved and Apply rates for OpenHands and SWE-agent. <sup>†</sup>Results reported on swebench.com (Jimenez et al., 2024). "—" indicates that the result is not reported on the SWE-Bench website.

The scale differences in Figure 4 translate into a large performance gap. Even the strongest model gpt-5.4 resolves only around 25% of SWE-EVO instances, and matched-model comparisons show the same pattern: gpt-5.2 drops from 72.80% on SWE-bench Verified to 22.92% on SWE-





(a) GPT-series models.



(b) Open-source models.

Figure 5 | Failure mode distribution for SWE-agent with several model trajectories of unresolved instances. Each instance is automatically labeled using gpt-5-mini (OpenAI, 2025b) with categories from Table 8.

EVO. Performance still follows intuitive scaling trends across model families. One exception is glm-5/4p7, which perform much better with SWE-agent than OpenHands, suggesting scaffold sensitivity in addition to benchmark difficulty. Taken together, these results indicate that SWE-EVO is a more demanding and more realistic benchmark for end-to-end software-evolution capabilities.

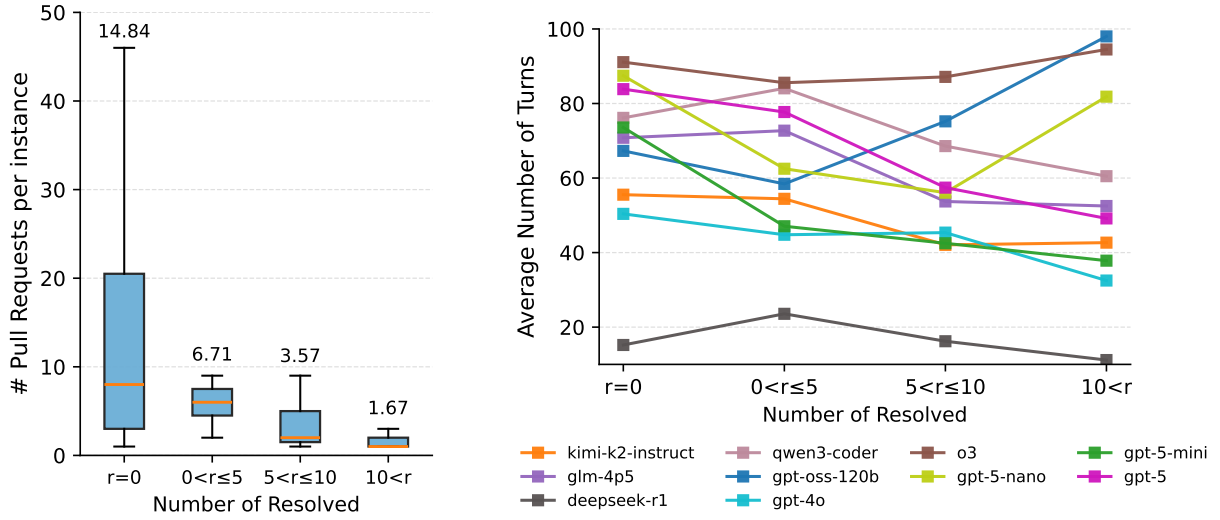
We also evaluate a more challenging *release-note only* setting, where agents receive no PR/issue context. As shown in Table 4 (in Appendix E), performance drops modestly but trends remain consistent, confirming that SWE-EVO poses substantial challenges regardless of specification detail.

**Uncertainty from Benchmark Scale.** Because SWE-EVO contains 48 curated instances, one resolved instance changes Resolved Rate by 2.08 percentage points. We therefore report uncertainty and avoid over-interpreting close leaderboard gaps. At the same time, the larger behavioral deltas in Figure 4 make each instance closer to a release-sized evolution task than to a single-issue repair task. Representative 95% Wilson confidence intervals over task instances are [14.9, 38.8] for 25.00%, [10.2, 31.9] for 18.75%, [4.5, 22.2] for 10.42%, and [0.4, 10.9] for 2.08%, so we focus on large performance gaps rather than small ranking differences.

**Fine-Grained Analysis with Fix Rate.** Beyond the binary *Resolved Rate*, we examine our soft metric *Fix Rate* (Section 3.2), which captures partial progress on large test suites. As shown in Table 6 (Appendix I), models indistinguishable under Resolved Rate can differ meaningfully: e.g., under OpenHands, both gpt-4.1 and gpt-oss-120b resolve only 2.08% of SWE-EVO, but their Fix Rates are 4.65% vs. 2.08%, revealing that gpt-4.1 repairs more failing tests per instance.

### 4.3. Analysis of Agent Behaviour

Beyond aggregate metrics, we investigate *why* agents fail on SWE-EVO by analyzing their execution trajectories. This qualitative analysis reveals distinct failure patterns across different model families.



(a) Average number of pull requests per difficulty group. (b) Average number of turns by model across difficulty groups.

Figure 6 | Difficulty analysis of SWE-EVO instances: (a) instances associated with more pull requests are harder; (b) stronger models scale turn usage with difficulty, while others show limited adaptivity.

#### 4.3.1. Trajectory Failure Modes Analysis

We perform an LLM-as-a-judge analysis of unresolved SWE-agent trajectories, following prior work on SWE-agent (Yang et al., 2024a). The judge assigns one primary label from the taxonomy in Table 8; these labels are qualitative diagnostics rather than execution-based scores, and human agreement validation remains future work. The taxonomy separates low-level execution failures from semantic failures: *Syntax Error* and *Tool-Use* capture invalid patches or failed interaction with the agent tools; *Incorrect Implementation* and *Instruction Following* capture wrong logic or misread requirements; *Stuck in Loop* and *Gave Up Prematurely* capture unproductive or prematurely terminated trajectories; and *Other* covers rare ambiguous cases.

**Results.** Figure 5 suggests distinct failure patterns across model families. For the GPT series, gpt-5 rarely fails due to syntax or tool issues; instead, over 60% of failures come from *Instruction Following*, suggesting difficulty interpreting complex release notes. Smaller variants (gpt-5-mini, gpt-5-nano) show increasing *Incorrect Implementation*, *Tool-Use*, and *Syntax Error* failures, while older models (o3, gpt-4.1, gpt-4o) exhibit more looping and early-termination issues. In contrast, open and hybrid models follow different patterns: kimi-k2-instruct, qwen3-coder, and gpt-oss-120b fail mainly due to *Incorrect Implementation*, indicating weaker semantic reasoning but stable tool use, whereas deepseek-r1 often gets stuck in loops or fails in execution, and glm-4p5 shows more evenly distributed errors across categories.

#### 4.3.2. Difficulty Analysis

A key property of SWE-EVO is that instance difficulty varies widely across multiple axes: PR count, specification length, files and lines edited, functions edited, and test-suite size (Table 1). Figure 6 groups instances by empirical difficulty, using the number of model-scaffold combinations that resolve each instance:  $r = 0$ ,  $0 < r \leq 5$ ,  $5 < r \leq 10$ , and  $r > 10$ , covering roughly 64%, 15%, 15%, and 6% of instances. Panel (a) shows that harder groups have substantially more linked PRs: unresolved-by-all

instances average 14.84 PRs, compared with 1.67 for the easiest group. This supports PR count as a useful proxy for release-level complexity. Panel (b) shows that stronger models tend to spend more turns on harder instances and terminate earlier on easier ones, while weaker models show less adaptive turn allocation.

## 5. Conclusion

We introduce SWE-EVO, a benchmark for evaluating coding agents on realistic software evolution rather than isolated bug fixing. SWE-EVO requires interpreting release-note-centered specifications, performing multi-file changes, and preserving functionality under comprehensive tests, making it substantially more challenging than SWE-Bench. Experiments with OpenHands and SWE-agent across 18 models reveal a large capability gap: the best model (gpt-5.4) solves only 25% of tasks, while a matched model such as gpt-5.2 drops from 72.80% on SWE-Bench Verified to 22.92% on SWE-EVO. An exploratory failure analysis suggests that stronger models struggle more with instruction following, while weaker ones exhibit tool-use and syntax errors. We expect SWE-EVO to complement existing benchmarks with a focus on long-horizon software evolution.

## 6. Limitations

Our work has some limitations. First, SWE-EVO currently covers Python library projects only; this choice keeps the benchmark reproducible and compatible with SWE-Bench/SWE-Gym environments, but multilingual and downstream-application evolution remain future work. Second, our task specifications are release-note-centered, optionally augmented with linked PR/issue text, and therefore do not cover all evolution drivers, such as security advisories, dependency updates, or design-document-driven refactors. Third, the benchmark has 48 curated instances and an imbalanced repository distribution, especially toward dvc. We deliberately prioritize large, execution-validated release transitions over a larger number of shallow tasks, but the current size still limits statistical power for fine-grained comparisons; we therefore provide repository-composition details in Appendix H. Finally, Fix Rate is execution-verifiable but still weights tests equally and does not measure code quality, maintainability, or patch minimality. We plan to release task metadata, construction scripts, Docker execution settings, generated patches, and aggregate result files to support reproducibility.

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## A. Evaluated Models

We evaluate SWE-EVO on a diverse set of recent LLMs from multiple providers, including both closed- and open-weight models. Table 3 lists all evaluated models.

| Provider    | Model                 | Type          | Reference                            |
|-------------|-----------------------|---------------|--------------------------------------|
| OpenAI      | gpt-5.4               | Closed-weight | (OpenAI, 2026)                       |
|             | gpt-5.2               | Closed-weight | (OpenAI, 2025c)                      |
|             | gpt-5-08-07           | Closed-weight | (OpenAI, 2025d)                      |
|             | gpt-5-mini-08-07      | Closed-weight | (OpenAI, 2025d)                      |
|             | gpt-5-nano-08-07      | Closed-weight | (OpenAI, 2025d)                      |
|             | o3-2025-04-16         | Closed-weight | (OpenAI, 2025f)                      |
|             | gpt-4.1-2025-04-14    | Closed-weight | (OpenAI, 2025a)                      |
|             | gpt-4o-2024-11-20     | Closed-weight | (OpenAI, 2024a)                      |
|             | gpt-oss-120b          | Open-weight   | (OpenAI, 2025e)                      |
| DeepSeek    | deepseek-v3p2         | Open-weight   | (DeepSeek-AI, 2025c)                 |
|             | deepseek-v3p1         | Open-weight   | (DeepSeek-AI, 2025b)                 |
|             | deepseek-r1-0528      | Open-weight   | (DeepSeek-AI, 2025a)                 |
| Zhipu AI    | glm-5                 | Open-weight   | (Z.ai Team, 2026)                    |
|             | glm-4p7               | Open-weight   | (Z.ai, 2025)                         |
|             | glm-4p5               | Open-weight   | (GLM-4.5 Team, 2025)                 |
| Qwen        | qwen3-coder-480b-a35b | Open-weight   | (Qwen Team, 2025; Yang et al., 2025) |
| Moonshot AI | kimi-k2p5             | Open-weight   | (Kimi Team, 2026)                    |
|             | kimi-k2-instruct      | Open-weight   | (Kimi Team, 2025a)                   |

Table 3 | Full list of LLMs evaluated on SWE-EVO, grouped by provider.

## B. Software Evolution Conceptual Model

Software evolution is an iterative cycle where, starting from an existing system, engineers identify required changes, analyze their impact, and carry out an evolution process that yields a new system aligned with updated requirements. This cycle repeats as the new system becomes the starting point for subsequent iterations. SWE-EVO captures this process by tasking agents with evolving codebases between consecutive release versions.

## C. Self-Evolving Agents and Tool Creation

Due to the huge design space for software agents, building an optimal agent scaffold can be extremely challenging and costly. As a result, several self-improving and self-evolving software agents have emerged recently. The Self-Improving Coding Agent (SICA) (Robeyns et al., 2025) introduced mechanisms for agents to learn from their own experiences. Darwin-Gödel Machine (DGM) (Zhang et al., 2025a) proposed open-ended evolution of self-improving agents through iterative refinement. Huxley-Gödel Machine (HGM) (Wang et al., 2025) further advanced this paradigm by approximating optimal self-improving machines for human-level coding. However, such self-improving agents typically require costly offline training on known benchmarks and may not generalize well across different LLMs, benchmarks, and issue types.



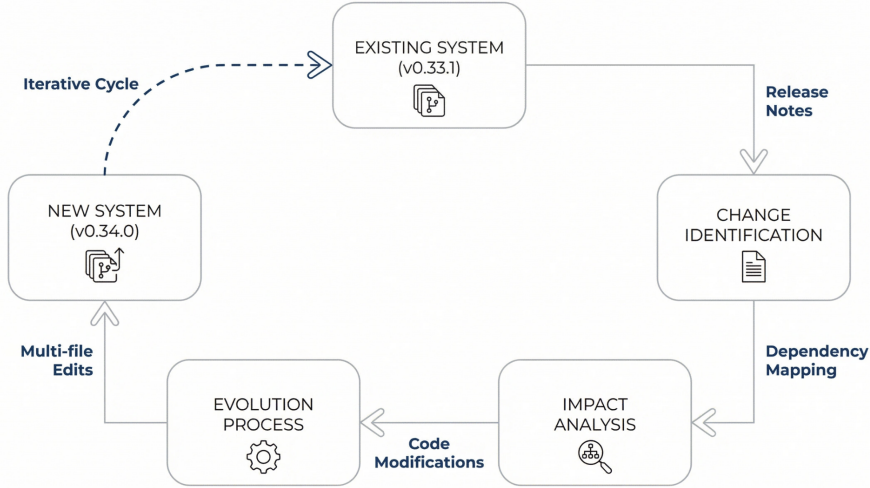


Figure 7 | Conceptual model of software evolution, depicting the iterative cycle from an existing system to a new system through change identification, impact analysis, and the evolution process.

Beyond the software engineering domain, prior work has explored using LLMs to create tools for general reasoning or embodied tasks. Large Language Models as Tool Makers (LATM) (Cai et al., 2024) demonstrated that LLMs can autonomously create reusable tools to solve complex tasks more efficiently. Voyager (Wang et al., 2024a) introduced an open-ended embodied agent that continuously acquires new skills through code generation in Minecraft. CREATOR (Qian et al., 2024) proposed disentangling abstract and concrete reasoning through tool creation. TroVE (Wang et al., 2024e) focused on inducing verifiable and efficient toolboxes for programmatic tasks. Alita (Qiu et al., 2025) presented a generalist agent enabling scalable agentic reasoning with minimal predefinition and maximal self-evolution. While these works demonstrate the potential of self-evolving agents and tool creation, they do not specifically target real-world software engineering problems that require long-horizon evolution across multiple commits and versions.

## D. SWE-EVO vs. SWE-Bench Comparison

The main paper reports the quantitative comparison in Figure 4. The same statistics are derived from instance-level metadata for SWE-EVO and SWE-Bench, covering specification length, gold-patch scope, and test-suite size.

## E. Context Comparison Results

We further compare performance under *release-note only* and *release-note + PR/issue context* settings. As shown in Table 4, adding PR/issue context generally improves resolved rates across most models, while preserving the overall ranking and relative trends. This suggests that additional context provides useful signals.

## F. Pull Request Distribution Analysis

Unlike SWE-Bench, where each task instance corresponds to a single pull request (PR), an instance in SWE-EVO may be associated with multiple pull requests that collectively implement or refine a release-note change. We hypothesize that instances associated with a larger number of pull requests

| Model                 | release-note only |              | release-note + PR/issue context |              |
|-----------------------|-------------------|--------------|---------------------------------|--------------|
|                       | OpenHands         | SWE-agent    | OpenHands                       | SWE-agent    |
| gpt-5-08-07           | <b>14.58</b>      | <b>16.67</b> | <b>18.75</b>                    | <b>20.83</b> |
| gpt-5-mini-08-07      | 8.33              | 10.42        | 10.42                           | 10.42        |
| o3-2025-04-16         | 4.17              | 6.25         | 4.17                            | 6.25         |
| gpt-4.1-2025-04-14    | 2.08              | 8.33         | 2.08                            | 10.42        |
| gpt-4o-2024-11-20     | 8.33              | 4.17         | 6.25                            | 6.25         |
| gpt-oss-120b          | 0.00              | 0.00         | 2.08                            | 6.25         |
| deepseek-r1-0528      | 10.42             | 6.25         | 10.42                           | 8.33         |
| glm-4p5               | 6.25              | 12.50        | 16.67                           | 16.67        |
| qwen3-coder-480b-a35b | <b>14.58</b>      | 12.50        | 14.58                           | 14.58        |
| kimi-k2-instruct      | 8.33              | 12.50        | 16.67                           | 18.75        |

Table 4 | Results on SWE-EVO under two settings: **release-note only** and **release-note + PR/issue context**. Reported metric is **Resolved Rate (%)** on **OpenHands** and **SWE-agent**.

reflect more complex, multi-step development efforts, as each pull request typically represents a distinct feature enhancement or bug fix. As shown in Figure 8, the number of pull requests per instance varies widely, indicating that SWE-EVO spans a broad range of difficulty levels.

## G. Difficulty Analysis

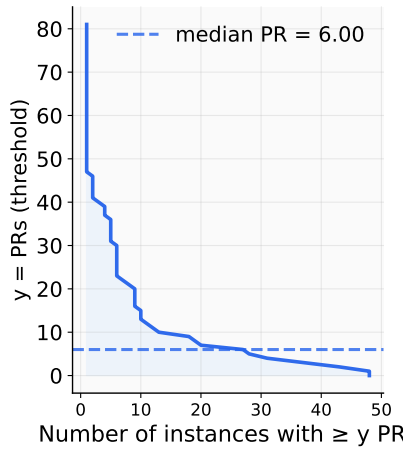
The main paper reports the difficulty analysis in Figure 6 and Section 4.3.2. We keep this appendix section as a pointer so the appendix PR-distribution analysis remains connected to the promoted main-text result without duplicating the same figure.

## H. Repository Composition and Bias Checks

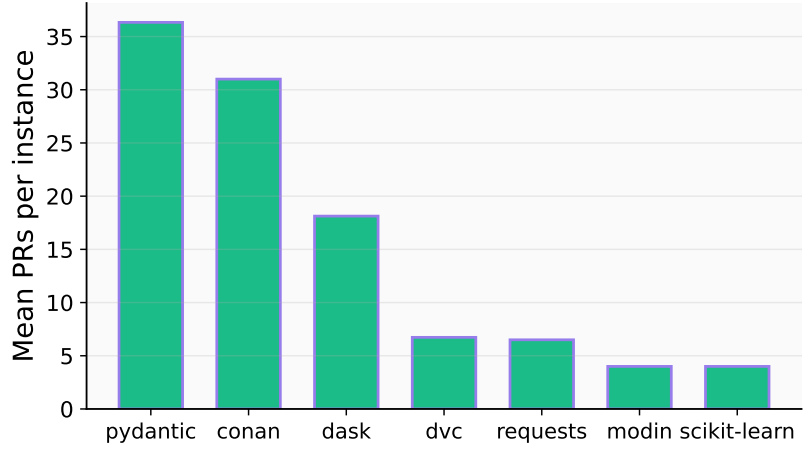
Table 5 reports per-repository Resolved Rate and Fix Rate, averaged over these evaluated models: gpt-5-08-07, gpt-5-mini-08-07, gpt-5-nano-08-07, o3-2025-04-16, gpt-4.1, gpt-4o, gpt-oss-120b, deepseek-r1, glm-4p5, qwen3-coder-480b, and kimi-k2-instruct. Our strict collection process requires stable release tags, reproducible environments, and executable tests, which improves validity but introduces some quantity imbalance: 26 of 48 instances come from dvc. However, the resulting metrics are not concentrated on a single repository. Performance remains low across repositories, and the highest aggregate values vary by scaffold. We therefore treat repository composition as a dataset caveat, but the metric itself does not appear to favor any single repository.

## I. Fix Rate Results

Beyond the binary *Resolved Rate*, we also examine our soft metric *Fix Rate*, which captures partial progress on large test suites (Section 3.2). As shown in Table 6, models that appear similar under Resolved Rate can differ meaningfully once partial improvements are considered. For example, under OpenHands, both gpt-4.1 and gpt-oss-120b resolve only 2.08% of SWE-EVO, but their Fix Rates differ (4.65% vs. 2.08%), indicating that gpt-4.1 consistently fixes more failing tests per instance. More broadly, we observe a consistent gap between Fix Rate and Resolved Rate across all models, suggesting that many trajectories make partial but incomplete progress. This finer granularity is



(a) Complementary cumulative distribution of PR counts, showing how many instances contain at least  $y$  linked pull requests.



(b) Mean PRs per repository, highlighting that certain codebases require substantially more upstream context to resolve.

Figure 8 | Pull Request statistics across SWE-EVO instances.

| Repository   | # Inst.   | Share (%)    | OpenHands |      | SWE-agent |      |
|--------------|-----------|--------------|-----------|------|-----------|------|
|              |           |              | Resolved  | Fix  | Resolved  | Fix  |
| dvc          | 26        | 54.2         | 4.36      | 6.69 | 8.72      | 5.63 |
| dask         | 8         | 16.7         | 2.68      | 1.14 | 2.57      | 0.83 |
| requests     | 4         | 8.3          | 2.65      | 2.46 | 3.12      | 2.92 |
| pydantic     | 3         | 6.2          | 0.00      | 0.00 | 0.01      | 0.00 |
| modin        | 3         | 6.2          | 1.23      | 1.14 | 1.04      | 1.04 |
| conan        | 2         | 4.2          | 0.38      | 0.38 | 0.31      | 0.21 |
| scikit-learn | 2         | 4.2          | 0.19      | 0.19 | 0.00      | 0.00 |
| <b>Total</b> | <b>48</b> | <b>100.0</b> | –         | –    | –         | –    |

Table 5 | Repository composition and aggregate per-repository performance. Resolved and Fix values are percentages averaged over these evaluated models for each scaffold.

especially important on SWE-EVO, where each task involves many tests, and binary outcomes would otherwise obscure systematic differences in model capability.

## J. FAIL\_TO\_PASS and PASS\_TO\_PASS Diagnostics

Table 7 reports detailed FAIL\_TO\_PASS and PASS\_TO\_PASS rates for each model in these evaluated model and each scaffold. These diagnostics separate two effects that are combined by strict Fix Rate: whether the model implements the requested behavioral change, and whether it preserves tests that already passed before the patch. Across most models, higher PASS\_TO\_PASS rates tend to coincide with lower FAIL\_TO\_PASS rates, suggesting a trade-off: as a model spends more effort resolving the new task, it may be more likely to forget or break existing codebase behavior. However, valid software evolution should satisfy the requested change without breaking existing behavior. We therefore use these rates as diagnostics, while keeping strict Fix Rate as the main soft metric.

| Model       |                       | OpenHands    |         | SWE-agent    |         | Average      |         |
|-------------|-----------------------|--------------|---------|--------------|---------|--------------|---------|
|             |                       | Resolved (%) | Fix (%) | Resolved (%) | Fix (%) | Resolved (%) | Fix (%) |
| OpenAI      | gpt-5.4               | 25.00        | 33.89   | 25.00        | 33.98   | 25.00        | 33.96   |
|             | gpt-5.2               | 18.75        | 23.43   | 22.92        | 30.21   | 20.84        | 26.82   |
|             | gpt-5-08-07           | 18.75        | 27.64   | 20.83        | 31.46   | 19.79        | 29.55   |
|             | gpt-5-mini-08-07      | 10.42        | 17.48   | 10.42        | 17.48   | 10.42        | 17.48   |
|             | gpt-5-nano-08-07      | 4.17         | 5.99    | 4.17         | 5.26    | 4.17         | 5.63    |
|             | o3-2025-04-16         | 4.17         | 6.47    | 4.17         | 13.72   | 4.17         | 10.10   |
|             | gpt-4.1-2025-04-14    | 2.08         | 4.65    | 10.42        | 14.79   | 6.25         | 9.72    |
|             | gpt-4o-2024-11-20     | 6.25         | 7.77    | 6.25         | 10.15   | 6.25         | 8.96    |
|             | gpt-oss-120b          | 2.08         | 2.08    | 6.25         | 7.88    | 4.17         | 4.98    |
| DeepSeek    | deepseek-v3p2         | 20.83        | 26.89   | 23.40        | 31.41   | 22.12        | 29.15   |
|             | deepseek-v3p1         | 16.67        | 21.83   | 10.42        | 15.51   | 13.55        | 18.67   |
|             | deepseek-r1-0528      | 10.42        | 14.31   | 8.33         | 9.89    | 9.38         | 12.10   |
| Zhipu AI    | glm-5                 | 8.33         | 9.67    | 37.50        | 44.27   | 22.92        | 27.45   |
|             | glm-4p7               | 4.17         | 5.19    | 39.58        | 45.87   | 21.88        | 25.53   |
|             | glm-4p5               | 16.67        | 23.74   | 16.67        | 26.55   | 16.67        | 25.15   |
| Qwen        | qwen3-coder-480b-a35b | 14.58        | 19.56   | 14.58        | 23.74   | 14.58        | 21.65   |
| Moonshot AI | kimi-k2p5             | 22.92        | 27.75   | 25.00        | 32.91   | 23.96        | 30.33   |
|             | kimi-k2-instruct      | 16.67        | 22.42   | 18.75        | 24.03   | 17.71        | 23.23   |

Table 6 | Comparison of **Resolved Rate (%)** and **Fix Rate (%)** across models on **OpenHands**, **SWE-agent**, and their **Average**.

## K. Data and Code Availability

SWE-EVO is designed for public release. We will release the task metadata, release-note and PR/issue context construction scripts, Docker-based execution configuration, evaluation harness, generated patches, and aggregate result files needed to reproduce the reported metrics. The release will also include the dataset fields described in Table 9 and instructions for running OpenHands and SWE-agent under the settings used in this paper. When required for anonymous review, these artifacts should be distributed through an anonymized repository or supplemental archive.

## L. Failure Mode Descriptions

We use a coarse taxonomy that separates execution failures, semantic failures, and unproductive trajectories. For each unresolved SWE-agent trajectory, the judge assigns exactly one primary label; when multiple issues are present, it chooses the most fundamental cause. Table 8 summarizes the labels used in the main analysis.

## M. Dataset Fields

Table 9 describes the SWE-EVO dataset fields and outlines how they are obtained throughout the curation process.

| Model            | OpenHands    |              | SWE-agent    |              |
|------------------|--------------|--------------|--------------|--------------|
|                  | FAIL_TO_PASS | PASS_TO_PASS | FAIL_TO_PASS | PASS_TO_PASS |
| gpt-5-08-07      | 65.68        | 34.32        | 59.95        | 37.97        |
| gpt-5-mini-08-07 | 64.15        | 23.35        | 76.27        | 21.65        |
| gpt-5-nano-08-07 | 58.59        | 8.07         | 90.58        | 7.34         |
| o3-2025-04-16    | 85.20        | 8.55         | 82.11        | 15.81        |
| gpt-4.1          | 82.85        | 4.65         | 71.67        | 24.16        |
| gpt-4o           | 87.20        | 10.89        | 81.62        | 16.30        |
| gpt-oss-120b     | 95.83        | 4.17         | 86.91        | 11.00        |
| deepseek-r1      | 77.32        | 22.68        | 83.23        | 14.69        |
| glm-4p5          | 68.64        | 29.28        | 62.57        | 35.35        |
| qwen3-coder-480b | 68.54        | 29.38        | 62.97        | 32.86        |
| kimi-k2-instruct | 71.45        | 28.55        | 67.52        | 30.39        |

Table 7 | FAIL\_TO\_PASS and PASS\_TO\_PASS rates (%) by model and scaffold. The diagnostics separate progress on required behavior from preservation of previously passing tests.

| Category                 | Description                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Syntax Error             | Patch introduces parse, formatting, import, indentation, JSON/YAML, or similar errors that prevent execution.         |
| Incorrect Implementation | Patch touches a plausible area but implements the required behavior incorrectly or incompletely.                      |
| Instruction Following    | Agent misreads, ignores, or deviates from the release note or linked requirement, effectively solving the wrong task. |
| Tool-Use                 | Progress is blocked by failed or incorrect tool use, such as bad edits, missed tests, or wrong paths.                 |
| Stuck in Loop            | Agent repeatedly reads, edits, or reruns tests without substantive progress.                                          |
| Gave Up Prematurely      | Agent stops or declares failure while reasonable next steps remain.                                                   |
| Other                    | Rare or ambiguous failure not captured by the other labels.                                                           |

Table 8 | Descriptions of failure mode categories.

## N. Problem Statement

For each SWE-EVO instance, the problem statement is defined by the official release notes text that describes the changes between the *start* version and the *end* version of this instance, as published by the project maintainers. This release note is then presented to the agent as the sole natural-language specification of how the codebase should change between the two versions.

### N.1. Pull Request, Issue Context

Release notes frequently refer to specific pull requests or issues (e.g., “see #1234” or “thanks to PR #5678”). To make these references actionable for agent, we augment the problem statement with the context of the linked artifacts. For each referenced pull request or issue, we fetch its body from github. These texts are then concatenated into a compact “PR / Issue Context” block that is presented to the agent alongside the release note. We then append these texts directly below the release note in a structured format, without rewriting or summarizing them. In general, the final problem statement presented to the agent has the form: *release-note* followed by a sequence of sections



| Field              | Type      | Description                                                                                                                                                                        |
|--------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| repo               | str       | Git repository identifier for the task instance, e.g., the GitHub owner/name.                                                                                                      |
| base_commit        | str       | The commit on which the pull request is based, representing the repository state before the issue is resolved.                                                                     |
| start_version      | str       | Git tag or version identifier of the starting release for this task instance.                                                                                                      |
| end_version        | str       | Git tag or version identifier of the target release after the change has been applied.                                                                                             |
| end_version_commit | str       | Git commit hash corresponding to the end_version tag.                                                                                                                              |
| patch              | str       | Gold patch proposed by the pull request, in .diff format.                                                                                                                          |
| test_patch         | str       | Modifications to the test suite proposed by the pull request that are typically used to check whether the issue has been resolved.                                                 |
| problem_statement  | str       | Issue description text, typically describing the bug or requested feature, used as the task problem statement.                                                                     |
| FAIL_TO_PASS       | List[str] | Test cases that are expected to successfully transition from failing to passing and are used to evaluate the correctness of the patch.                                             |
| PASS_TO_PASS       | List[str] | Test cases that are already passing prior to applying the gold patch; a correct patch should not introduce regression failures in these tests.                                     |
| *image             | str       | Instance-level Docker image that provides an execution environment.                                                                                                                |
| *test_cmds         | List[str] | Command(s) used to run the test suite, as identified by the verify agent in REPOLAUNCH, enabling detailed logging of each test item's status (e.g., via <code>pytest -rA</code> ). |
| *log_parser        | str       | Type of log parser required for the instance—by default, <code>pytest</code> .                                                                                                     |

Table 9 | Required fields for a typical issue-solving task instance. Fields marked with \* are newly added compared to SWE-Bench.

\n### PR xxx:\n or \n### Issue yyy:\n with their corresponding content for every referenced pull request and issue. so that the agent sees the original release note followed by the full text of any referenced pull requests and issues.

## N.2. Example

To make this concrete, we present two example problem statements (Boxes N.2 and N.2), corresponding to the instances `iterative__dvc_0.33.1_0.34.0` in the `iterative/dvc` repository and `dask__dask_2023.6.1_2023.7.0` in the `dask/dask` repository.

**Problem Statement: iterative\_\_dvc\_0.33.1\_0.34.0**

1) ['dvc metrics show' now nicely formats multiline metrics files like tsv/htsv, csv/hcsv, json](https://github.com/iterative/dvc/issues/1716); Kudos @mroutis :medal\_sports:  
 2) ['dvc remote add' no longer silently overwrites existing sections](https://github.com/iterative/dvc/issues/1760);  
 3) [Use a workaround to bypass SIP protection on osx, when accessing libSystem](https://github.com/iterative/dvc/issues/1515);  
 4) [Don't try to create existing container on azure](https://github.com/iterative/dvc/issues/1811); Kudos @AmitAronovitch :medal\_sports:  
 5) Dvc repository now uses read-only http remote for our images instead of s3;  
 6) Fix bug in 'dvc status' where an error is raised if cache is not present locally nor on the remote; 7) [Fix progress bar on 'dvc pull']  
 (https://github.com/iterative/dvc/issues/1807); Kudos @pared :medal\_sports:  
 8) [Automatically detect metrics file type by extension](https://github.com/iterative/dvc/issues/1553); Kudos @cand126 :medal\_sports:

Welcome new contributor @AmitAronovitch ! :tada:

**### Issue 1716:**

When displaying TSV metrics output the printout is not readable:

```
../20181223-TrainSetZero/SanityCheck/Bravo_on_TrainSetZero.metrics:  value_mse
deviation_mse data_set
0.421601 0.173461 train
0.67528 0.289545 testing
0.671502 0.297848 validation
```

I think it would be much easier to read if a newline were added, and the remaining text were formatted as though it had been passed via column -t:

```
../20181223-TrainSetZero/SanityCheck/Bravo_on_TrainSetZero.metrics:
value_mse deviation_mse data_set
0.421601 0.173461 train
0.67528 0.289545 testing
0.671502 0.297848 validation
```

**### Issue 1807:**

```
(3.7.0-dvc) →dvc git:(dvctags) ×dvc pull
Preparing to download data from 's3://dvc-share/dvc/'
Preparing to collect status from s3://dvc-share/dvc/
[#####] 100% Collecting information
[#####] 100% Analysing status.
(1/3): [#####] 100% dvc_up.bmp
(2/3): [#####] 100% dvc.ico
(3/3): [#####] 100% dvc_left.bmp
(4/3): [#####] 100% Checkout finished!
looks like checkout didn't reset progress counter
```

**### Issue 1553:**

E.g. if we see that output has .json suffix, we could safely assume that it is -type json without explicitly specifying it.

**Problem Statement: iterative\_\_dvc\_0.33.1\_0.34.0 (continued)****### Issue 1760:**

The `dvc remote add` command ignores the existing remote and overwrites it silently.

```
→dvc -version
0.32.1+7d7ed4
```

To reproduce

```
dvc remote add s3 s3://bucket/subdir
dvc remote add s3 s3://bucket/subdir2
```

Expected behavior

The second command `dvc remote add s3 s3://bucket/subdir2` should fail with the Remote with name "s3" already exists message.

Current behavior

Remote URL is silently overwritten:

```
> cat .dvc/config
[ 'remote "s3"' ]
url = s3://bucket/subdir2
```

**### Issue 1811:**

Azure SAS connection strings can be used to allow read-only access to specific container, which is useful in the context of CI pipelines and automation.

However, when using such limited-credentials connection string, `dvc pull` abends with the error below.

Some digging reveals that this is because `remote/azure.py` always tries to create the bucket (which already exists). If we check for existence before trying to create - this should work...

(will send PR)

```
[##### ] 30% Collecting information
```

```
Client-Request-ID=19129d6a-5393-11e9-80ab-5800e34ea0d9
```

```
Retry policy did not allow for a retry:
```

```
Server-Timestamp=Sun, 31 Mar 2019 08:58:06 GMT,
Server-Request-ID=c72fbeda-501e-0089-3c9f-e78298000000,
HTTP status code=403, Exception=This request is not
authorized to perform this operation.
```

```
ErrorCode: AuthorizationFailure
```

```
<?xml version="1.0" encoding="utf-8"?><Error><Code>AuthorizationFailure</Code>
```

```
<Message>This request is not authorized to perform this operation.
```

```
RequestId:c72fbeda-501e-0089-3c9f-e78298000000
```

```
Time:2019-03-31T08:58:06.2059267Z</Message></Error>.
```

```
Error: failed to pull data from the cloud - This request is not authorized to perform this operation. ErrorCode: AuthorizationFailure
```

```
<?xml version="1.0" encoding="utf-8"?><Error><Code>AuthorizationFailure
```

```
</Code><Message>This request is not authorized to perform this operation.
```

```
RequestId:c72fbeda-501e-0089-3c9f-e78298000000
```

```
Time:2019-03-31T08:58:06.2059267Z</Message></Error>
```

Having any troubles? Hit us up at <https://dvc.org/support>, we are always happy to help!

Please provide information about your setup

DVC version: 0.29.0+220b4b.mod

linux x86\_64 py3.6 pip

**### Issue 1515:**

```
$ dvc -V
```

```
0.23.2+bad2ef.mod
```

DVC creates hardlinks instead of reflinks in APFS.

File system:

```
$ diskutil info / | grep -i system
```

```
File System Personality: APFS
```

**Problem Statement: iterative\_\_dvc\_0.33.1\_0.34.0 (continued)**

See last two lines:

```
$ dvc add Tags.xml -v
Debug: PRAGMA user_version;
Debug: fetched: [(3,)]
Debug: CREATE TABLE IF NOT EXISTS state (inode INTEGER PRIMARY KEY, mtime TEXT NOT NULL, size
TEXT NOT NULL, md5 TEXT NOT NULL, timestamp TEXT NOT NULL)
Debug: CREATE TABLE IF NOT EXISTS state_info (count INTEGER)
Debug: CREATE TABLE IF NOT EXISTS link_state (path TEXT PRIMARY KEY, inode INTEGER NOT NULL,
mtime TEXT NOT NULL)
Debug: INSERT OR IGNORE INTO state_info (count) SELECT 0 WHERE NOT EXISTS (SELECT * FROM
state_info)
Debug: PRAGMA user_version = 3;
Debug: Skipping copying for '/Users/dmitry/src/modules-example/tmp/
test/Tags.xml', since it is not a symlink or a hardlink.
Adding 'Tags.xml' to '.gitignore'.
Saving 'Tags.xml' to cache '.dvc/cache'.
Debug: Path /Users/dmitry/src/modules-example/tmp/test/Tags.xml inode 12888021464
Debug: SELECT * from state WHERE inode=12888021464
Debug: fetched: []
Debug: INSERT INTO state(inode, mtime, size, md5, timestamp) VALUES (12888021464,
"1547854167848187904", "0", "6d861a1605e60b2f3a977a5a2a4419a2", "1547854178609126912")
Debug: File '/Users/dmitry/src/modules-example/tmp/test/.dvc/cache/6d/
861a1605e60b2f3a977a5a2a4419a2', md5 '6d861a1605e60b2f3a977a5a2a4419a2', actual 'None'
Debug: Cache type 'reflink' is not supported: reflink is not supported
Debug: Created 'hardlink': /Users/dmitry/src/modules-example/tmp/
test/.dvc/cache/6d/861a1605e60b2f3a977a5a2a4419a2 ->/Users/dmitry/src/
modules-example/tmp/test/Tags.xml
```

**Problem Statement: dask\_\_dask\_2023.6.1\_2023.7.0****2023.7.0**

Released on July 7, 2023

**Enhancements**

- Catch exceptions when attempting to load CLI entry points (#10380) Jacob Tomlinson

**Bug Fixes**

- Fix typo in `_clean_ipython_traceback` (#10385) Alexander Clausen
- Ensure that `df` is immutable after `from_pandas` (#10383) Patrick Hoefler
- Warn consistently for `inplace` in `Series.rename` (#10313) Patrick Hoefler

**Documentation**

- Add clarification about output shape and reshaping in `rechunk` documentation (#10377) Swayam Patil

**Maintenance**

- Simplify `astype` implementation (#10393) Patrick Hoefler
- Fix `test_first_and_last` to accommodate deprecated `last` (#10373) James Bourbeau
- Add level to `create_merge_tree` (#10391) Patrick Hoefler
- Do not derive from `scipy.stats.chisquare` docstring (#10382) Doug Davis

**### PR 10385:**

Regression from (#10354) <https://api.github.com/repos/dask/dask/pull/10354> - exception types of unhandled exceptions in ipython contexts were mangled and always set to `type`.

To reproduce, run the following in a jupyter notebook cell with dask 2023.6.1:

```
import dask
raise TypeError("wat")
```

Then, observe the websocket connection to jupyter; the message with `msg_type="error"` looks like this:

[\(Image Link\)](#)

Notice `content["ename"]` is set to `"type"`.

With dask 2023.6.0, `content["ename"]` is properly set to `"TypeError"`:

[\(Image Link\)](#)

These values aren't inconsequential - they end up being serialized into the ipynb file, and cause hard to debug issues.

(nb: I would prefer if this kind of issue was not something that can be caused by `import dask`, by opting in to this functionality by configuration or actively registering the hook functions)

- Tests added / passed
- Passes `pre-commit run --all-files`

Thanks to [\(@matbryan52\)](#) for the help debugging this.





**Problem Statement: dask\_\_dask\_2023.6.1\_2023.7.0 (continued)****### PR 10383:**

- Tests added / passed
- Passes `pre-commit run --all-files`

We had a similar problem in dask-expr

**### PR 10393:**

- Tests added / passed
- Passes `pre-commit run --all-files`

**### PR 10391:**

- Passes `pre-commit run --all-files`

**### PR 10373:**

This fixes

FAILED dask/dataframe/tests/test\_dataframe.py::  
test\_first\_and\_last[last] - FutureWarning: last is deprecated  
and will be removed in a future version. Please create a mask  
and filter using `:loc` instead  
which we're currently seeing in the upstream build (xref (#10347) ([Workflow Run URL](#)))

Python 3.10 Test Summary

dask/dataframe/tests/

test\_multi.py::test\_concat\_categorical[True-False-True]:

[XPASS(strict)] fails on pandas dev:

<https://api.github.com/repos/dask/dask/issues/10558>

dask/dataframe/tests/

test\_multi.py::test\_concat\_categorical[False-False-True]:

[XPASS(strict)] fails on pandas dev:

<https://api.github.com/repos/dask/dask/issues/10558>

Note that we were already emitting a FutureWarning for `DataFrame.last`, so this is a tests-only PR

cc (@j-bennet) (@phoff)

**### PR 10382:**

- Closes (#10381) Looks like our docs are running into an error on main. See this build from a recent PR <https://readthedocs.org/projects/dask/builds/21146384/>

x Tests added / passed

- Passes `pre-commit run --all-files`

`scipy.stats.chisquare` recently started using a doi Sphinx directive that exists in their repo but it's not importable. This is breaking dask docs generation. We can just point folks to that docstring instead of deriving from it.

for reference here is the PR in scipy: ([scipy/scipy#17682](#))

**### PR 10377:**

Closes (#10361) It is not documented anywhere that `dask.array.rechunk` expects the output to have the same shape as the input and does not allow reshaping. The validation step is also quite hidden. This has led to some confusion ([dask/distributed#7897 \(comment\)](#)), so I think it would be good to add a remark about this in the documentation.

- Tests added / passed
- Passes `pre-commit run --all-files`

**### PR 10313:**

The previous warning seemed very inconsistent

**### PR 10380:**

- Closes (#10379) If a package registers an entry point for the CLI but the entrypoint is broken or not importable the CLI cannot work at all.

This is a little hard to create an MRE for but I seeing this with `dask-ctl` when updating textual to a more recent version. The error comes from the following line.

([dask/dask/cli.py](#))

Line 97 in /dask/dask/commit/85c99bc20abc382774cfb6e5bf5f2db76ac09378

`command = entry_point.load()`

When this line is called third-party code is loaded, so we have no control over what happens. We could broadly catch exceptions here, print a warning and carry on.

- Tests added / passed
- Passes `pre-commit run --all-files`